Resolve Tinea Cream

Miconazole nitrate

This leaflet answers some common questions about Resolve Tinea cream

Description:

ResolveTinea Cream is a white to off-white glossy, firm, thick cream.

Constituents:

ResolveTinea Cream contains miconazole nitrate 2% w/w as the active ingredient and phenethyl alcohol 1% w/w as the preservative.

Action:

Pharmacodynamics
Resolve Tinea Cream contains miconazole nitrate 2 % w/w as the antifungal active ingredient in a smooth cream base suited to the treatment of sensitive inflamed skin and the skin of the flexural areas that is prone to maceration and chafing.

The base of the cream was formulated so that it would have a soothing effect with light liquid paraffin included to enhance the emollient properties and act as a moisturising agent to help treat and prevent dry skin.

Dimethicone has been included to provide protection against water, urine, sweat and other irritants and to give the skin a smoothness which helps prevent chafing. Helping to restore the barrier function to the skin is vital to the healing process.

Phenethyl alcohol is incorporated as a preservative. It was selected to complement the action of miconazole and increase the antibacterial activity of the product.

The combination of emulsifying and stabilising agents were chosen predominantly because they produced a stable emulsion at a suitable pH.

Pharmacotherapeutic Group of Miconazole

Resolve Tinea cream contains miconazole nitrate 2% w/w, an imidazole antifungal agent which has a broad spectrum of antifungal activity including dermatophytes, pathogenic yeasts and moulds, and is also active against Gram-positive bacteria.

Miconazole is particularly active against species of medical interest such as *Candida, Trichophyton, Epidermophyton, Microsporum, Pityrosporum,* other yeast-like fungi and dermatophytes as well as Gram positive bacteria such as *Streptococcus pyogenes* and *Staphylococcus aureus*.

It has been used extensively for chronic skin infections which have not responded satisfactorily to other agent. It has also been used for over twenty years in both human and veterinary medicine as an anti-fungal drug.

Mechanism of Action of Miconazole

Miconazole is an antifungal agent and, like other imidazole derivatives, acts by altering the permeability of the cell membrane in sensitive fungi.

Minimum inhibitory concentrations for most fungi typically vary between 0.1 and 10 µg/mL.

There is much experimental evidence and it is now generally accepted that the major anti-fungal mechanism of the imidazoles is interference with ergosterol synthesis. This action is observed at low concentrations of imidazoles.

The *in vitro* Antifungal Activities of Miconazole and other Imidazoles

There have been a number of excellent studies on the *in vitro* antifungal activities of a range of imidazole derivatives.

The most extensive study is a comparison of miconazole with ketoconazole against 175 isolates of human fungal pathogens where the activity of ketoconazole against dermatophytic fungi was shown to be similar to that of miconazole.

A further study of five imidazole derivatives (clotrimazole, econazole, ketoconazole, miconazole and tioconazole) on the inhibition and killing of 11 isolates of *Candida albicans* (another yeast organism) has also been done. By spectrophotometric determination of 50% growth inhibition, all isolates were inhibited at low concentrations. In killing, using minimum fungicidal concentration, the overall similarity in efficacy of the range of imidazoles is clearly evident.

Another relevant study has evaluated the in vitro activity of five different antimycotics including miconazole, econazole and clotrimazole, against nineteen strains of *Pityrosporum orbiculare*. The results indicate that only minor differences were seen in the MIC of miconazole, clotrimazole and econazole. Three of the strains of *Pityrosporum orbiculare* were less susceptible to miconazole, clotrimazole and econazole. The patients from whom these strains were cultured were treated with miconazole cream and all were cured. The conclusion reached was that topical treatments allowed sufficiently high concentration of imidazole to provide clinical effect despite the high MIC observed *in vitro*.

Prolonged *in vitro* studies induced no resistance to Candida, Aspergillus or dermatophytes nor has this been observed in numerous clinical trials of the drug used topically or systemically.

On balance, the evidence comparing all imidazoles over a range of fungal species would indicate that the strong fungistatic effects are virtually indistinguishable, with little differences between imidazoles to promote any difference when used topically. However, imidazoles like ketoconazole which are water soluble are much better candidates for systemic therapy. The virtual non-absorption of miconazole when administered topically contributes to its safety in use.

Pharmacokinetics
The absorption of miconazole is not significant when applied to the skin or mucous membranes. Oral human administration indicates that miconazole has low bioavailability, plasma and urine levels are low and most of the drug can be recovered unmetabolised from the faeces.

Uses (Indications):

Effective treatment for tinea, such as athlete's foot, jock itch and ringworm. Also effective for skin infections such as pityriasis versicolour, thrush, thrush infected napkin rash, fungal infections where bacterial infection may be present. Protective silicone cream base helps to protect the skin from chafing, water, urine, other irritants.

Recommended dosage:

Clean and dry affected area thoroughly.
Apply Resolve Tinea Cream to the affected and surrounding skin twice daily. Continue treatment for 2 weeks after symptoms disappear to avoid recurrence. Cleanse with a soap alternative as soap may irritate the skin.

Mode of administration:

Topical.

Contraindications:

Known sensitivity to any of the ingredients.

Warnings and precautions:

Avoid contact with eyes. In patients on warfarin, caution should be exercised and the anticoagulant effect should be monitored (see Interactions).

Interactions with Other Medicaments:

Miconazole administered systemically is known to inhibit CYP2C9 enzyme system. Due to the limited systemic availability after topical application, clinically relevant interactions occur very rarely. In patients on warfarin which is subjected to metabolism by CYP2C9, caution should be exercised and the anticoagulant effect should be monitored (see Warnings and Special Precautions).

Pregnancy and lactation:

The safety of miconazole nitrate is well reported and, when used in topical medication, is a category A drug as defined in 'Medicines in Pregnancy' approved by the Australian Drug Evaluation Committee.

Category A drugs are those which have been taken by a large number of pregnant women and women of child-bearing age without any proven increase in the frequency of malformations or other direct or indirect harmful effects on the foetus having been observed.

Side effects:

Allergy to any of the ingredients may cause skin irritation.

Overdose:

Overdose of Resolve Tinea Cream not applicable as it is a topical preparation.

Storing:

Store below 30°C

Packs available:

25g tube.

Shelf life:

The shelf life of Resolve Tinea cream is 3 years.

Note:

This information is limited. For further information please consult your doctor or pharmacist.

- You may want to read this leaflet again, please do not throw it away until you have finished using your medicine.

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